

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0737

Stonebridge 138 kV
March 29, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0737. The Interconnection Customer proposes a 187.0 MW solar photovoltaic facility interconnecting at the Stonebridge 138 kV substation in the Virginia area. The request was received on January 31, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 214 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$19,558,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sunset / Saddlebrooke Ranch 138 kV	Easton 2 / Vienna Operations 345 kV	156.9	512	41.9
Vienna Operations 345-1 kV Transformer Bank	Easton 2 / Vienna Operations 345 kV L	154.4	552	49.5
Randall County / Chemolite 138 kV	Vienna Operations / Easton 2 345 kV	150.8	230	15.3
Tasley Substation 345/138kV XFMR	Commonwealth Chesapeake / Southtown 345 kV	148.0	344	69.0
Ripple 345/138kV Transformer Bank	Vienna Operations / Southtown 345 kV	124.6	433	35.2
Quince Orchard / Southtown 138 kV	Commonwealth Chesapeake / Southtown 345 kV Line	123.8	236	7.6
Newmark / Ess Lewes 138 kV	Easton 2 / Vienna Operations 345 kV	121.6	368	25.0
Galvanizing / Santeetlah 138 kV	Southtown / Vienna Operations 345 kV	120.8	520	22.0
Sunset / Santeetlah 138 kV	Southtown / Vienna Operations 345 kV	120.4	527	22.0
Sunset Substation / St Bonifacius 138 kV	Southtown Substation / Vienna Operations 345 kV	120.2	1,064	54.5
W -Station / St Bonifacius 138 kV	Southtown / Vienna Operations 345 kV	119.5	1,089	54.5
Prunedale / Dane 138 kV L	Commonwealth Chesapeake / Southtown 345 kV Line	116.6	226	5.6
Sandburg / Jarratt Substation 138 kV	Vienna Operations / Easton 2 345 kV	114.8	542	27.1
Sandburg / Sherwood 138 kV	Vienna Operations Substation / Easton 2 345 kV	114.2	554	27.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Hitchland / Sherwood 138 kV L	Vienna Operations / Easton 2 Substation 345 kV L	113.9	561	27.1
Cady / Easton 2 345 kV L	Ripple / Linde 138 kV	113.4	1,185	66.6
North Lake / Kiwi 138 kV	Easton 2 Substation / Vienna Operations 345 kV	113.2	528	24.6
Mesquite / Ripple 138 kV	Southtown Substation / Vienna Operations 345 kV L	113.0	824	45.4
Lammers / Chemolite Substation 138 kV	Vienna Operations / Easton 2 345 kV	113.0	318	15.3
Hitchland / N Lake 138 kV L	Easton 2 / Vienna Operations 345 kV	112.5	591	27.1
Mesquite / Arroyo Tap 138 kV	Southtown / Vienna Operations 345 kV Line	112.0	859	45.4
Ocean City / Reidland Substation 138 kV	Southtown / Vienna Operations Substation 345 kV	111.4	878	45.4
Arroyo Tap / Reidland 138 kV	Southtown / Vienna Operations 345 kV	111.4	878	45.4
Ocean City / Candelaria Arenas 1718 138 kV	Southtown / Vienna Operations 345 kV	111.0	893	45.4
Memphis Junction / Lammers 138 kV L	Vienna Operations / Easton 2 345 kV line	111.0	499	32.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sunset / Saddlebrooke Ranch 138 kV	Rebuild line with higher-capacity conductor	\$8,471,000	6% (\$550,000)
Vienna Operations 345-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$19,893,000	8% (\$1,601,000)
Randall County / Chemolite 138 kV	Rebuild line with higher-capacity conductor	\$7,862,000	6% (\$493,000)
Tasley Substation 345/138kV XFMR	Replace transformer with higher-rated unit	\$13,055,000	22% (\$2,888,000)
Riple 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$14,214,000	8% (\$1,102,000)
Quince Orchard / Southtown 138 kV	Rebuild line with higher-capacity conductor	\$3,106,000	4% (\$117,000)
Ess Lewes / Newmark 138 kV line	Rebuild line with higher-capacity conductor	\$11,034,000	7% (\$781,000)
Galvanizing / Santeetlah 138 kV line	Rebuild line with higher-capacity conductor	\$4,420,000	6% (\$271,000)
Sunset / Santeetlah 138 kV	Rebuild line with higher-capacity conductor	\$10,865,000	6% (\$666,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sunset Substation / St Bonifacius 138 kV	Rebuild line with higher-capacity conductor	\$3,689,000	6% (\$240,000)
West -Station / St Bonifacius 138 kV	Rebuild line with higher-capacity conductor	\$8,252,000	6% (\$537,000)
Prunedale / Dane 138 kV L	Rebuild line with higher-capacity conductor	\$2,917,000	4% (\$110,000)
Sandburg / Jarratt Substation 138 kV	Rebuild line with higher-capacity conductor	\$6,729,000	7% (\$450,000)
Sandburg / Sherwood 138 kV	Rebuild line with higher-capacity conductor	\$4,070,000	7% (\$272,000)
Hitchland / Sherwood 138 kV L	Rebuild line with higher-capacity conductor	\$7,388,000	7% (\$494,000)
Cady / Easton 2 345 kV L	Rebuild line with higher-capacity conductor	\$74,852,000	7% (\$5,206,000)
Kiwi / N Lake 138 kV	Rebuild line with higher-capacity conductor	\$7,377,000	7% (\$503,000)
Mesquite / Ripple 138 kV	Rebuild line with higher-capacity conductor	\$4,583,000	8% (\$348,000)
Chemolite / Lammers 138 kV	Rebuild line with higher-capacity conductor	\$7,778,000	6% (\$488,000)
Hitchland / N Lake 138 kV L	Rebuild line with higher-capacity conductor	\$7,636,000	7% (\$511,000)
Mesquite / Arroyo Tap 138 kV	Rebuild line with higher-capacity conductor	\$2,948,000	8% (\$224,000)
Ocean City / Reidland 138 kV	Rebuild line with higher-capacity conductor	\$3,643,000	8% (\$277,000)
Arroyo Tap / Reidland 138 kV	Rebuild line with higher-capacity conductor	\$741,000	8% (\$56,000)
Cadelaria Arenas 1718 / Ocean City 138 kV L	Rebuild line with higher-capacity conductor	\$3,208,000	8% (\$244,000)
Memphis Junction / Lammers 138 kV L	Rebuild line with higher-capacity conductor	\$17,870,000	6% (\$1,129,000)
	Total allocated to project		\$19,558,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.